AccessAble AI

Website Accessibility Review

AI-assisted visual screening. Findings require verification.

Accessibility Review Report: https://cloud.google.com/

Date: October 24, 2024

Website URL: https://cloud.google.com/

Page Title: AI and Cloud Computing Services | Google Cloud

Executive Summary

This report provides a preliminary accessibility review of the Google Cloud homepage. The review identified several potential accessibility barriers that could impact users of assistive technologies, such as screen readers.

The primary areas of concern are:

- **Missing Accessible Names:** Several interactive elements (buttons) lack descriptive names, making them difficult or impossible for screen reader users to navigate.

- **Non-Descriptive Alternative Text:** Some images use generic alt text that fails to convey necessary context or content.

- **Improper Heading Hierarchy:** The page's heading structure skips levels, which can disrupt the logical flow for users navigating by headings.

Note: This review is based on automated observations and visual inspection. To ensure full compliance with WCAG standards, manual testing with keyboard navigation and various screen readers (e.g., NVDA, JAWS, VoiceOver) is strongly recommended.

Detailed Findings

1. Missing Accessible Names for Interactive Elements

- **Issue:** Several buttons lack an accessible name (text content or `aria-label`), making them "unnamed" to assistive technologies.

- **Evidence:** Observations identified buttons with `text: 'menu'` and `ariaLabel: null`, as well as multiple buttons where both the text and the aria-label are empty/null.

- **Potential Impact:** Users who rely on screen readers will encounter these buttons and only hear "button" without any indication of their purpose (e.g., they won't know the menu button opens a navigation menu). This can make critical parts of the site unusable.

- **Priority:** **High**

- **Suggested Fix:** Ensure every interactive element has a discernible name. For icon-only buttons, provide a descriptive `aria-label` (e.g., `aria-label="Open main menu"`). For purely decorative buttons, use `aria-hidden="true"`.

- **How to Verify:** Use a screen reader to navigate to these buttons and ensure they announce a meaningful, descriptive name.

2. Non-Descriptive Alternative Text

- **Issue:** Several images use generic or non-descriptive alternative text.

- **Evidence:** Images were found with alt text such as `"AI generated image"`, `"icon"`, and `"bar image"`.

- **Potential Impact:** When a screen reader encounters `"AI generated image"`, it provides the *method* of creation rather than the *content* or *purpose*. This deprives visually impaired users of the intended context or information.

- **Priority:** **Medium**

- **Suggested Fix:** Update alt text to describe the actual content or purpose of the image. For example, instead of `"AI generated image"`, use a description of what is being depicted. If an image is purely decorative, use a null alt attribute (`alt=""`).

- **How to Verify:** Inspect image `alt` attributes in the code and perform a manual check with a screen reader to ensure descriptions are helpful and contextually relevant.

3. Improper Heading Hierarchy

- **Issue:** The heading structure skips heading levels, breaking the logical outline of the page.

- **Evidence:** The page uses an `H1` ("The new way to cloud") followed immediately by several `H3` tags (e.g., "Build with the cloud platform...", "Build and scale generative AI apps..."), skipping the `H2` level.

- **Potential Impact:** Screen reader users often navigate by jumping between headings to understand the page structure. Skipped levels can cause confusion regarding the relationships between content sections.

- **Priority:** **Medium**

- **Suggested Fix:** Reorganize HTML heading tags to follow a logical, nested order (e.g., `H1` $\rightarrow$ `H2` $\rightarrow$ `H3`).

- **How to Verify:** Use a browser extension that generates a heading outline or use a screen reader to navigate the heading structure to ensure it is sequential and nested correctly.

Recommendations & Next Steps

1. **Conduct Manual Testing:** Perform comprehensive testing using only a keyboard and various screen readers to identify issues that automated tools might miss (e.g., focus management, logical reading order).

2. **Audit All Interactive Elements:** Systematically check all buttons, links, and form inputs for accessible names and clear labels.

3. **Review Image Alt Text:** Perform a content audit of all visual elements to ensure alt text is descriptive and purposeful.

4. **Fix Heading Structure:** Implement a valid, nested heading hierarchy across all main pages.

5. **Continuous Monitoring:** Integrate accessibility testing into your regular development and deployment workflows.

Disclaimer: This report is a preliminary assessment and does not guarantee WCAG compliance.